

Contents



EDITOR : Ramesh Chopra

EDITORIAL CORRESPONDENCE : Editorial Secretary
Phone: 011-26810601; E-mail: editsec@efy.in
(Technical queries: efylab@efy.in)

SUBSCRIPTIONS & MISSING ISSUES : Phone: 011-26810601 or 02 or 03
E-mail: support@efy.in

BACK ISSUES, BOOKS, CDs, PCBs etc. : Kits'n'Spares, New Delhi
Phone: 011-26371661, 26371662
E-mail: info@kitsnspares.com

NEWSSTAND DISTRIBUTION : Ph: 011-40596600
E-mail: efycirc@efy.in

ADVERTISEMENTS NEW DELHI (HEAD OFFICE) : Ph: 011-26810601 or 02 or 03
E-mail: efyenq@efy.in

MUMBAI : Ph: 022-24950047, 24928520
E-mail: efymum@efy.in

BENGALURU : Ph: 080-25260394, 25260023
E-mail: efylbr@efy.in

PUNE : Ph: 08800295610, 09870682995
E-mail: efypune@efy.in

GUJARAT : Ph: 079-61344948
E-mail: efyahd@efy.in

JAPAN : Tandem Inc., Ph: 81-3-3541-4166
E-mail: japan@efy.in

SINGAPORE : Publicitas Singapore Pte Ltd
Ph: +65-6836 2272
E-mail: singapore@efy.in

TAIWAN : J.K. Media, Ph: 886-2-87726780 ext. 10
E-mail: taiwan@efy.in

UNITED STATES : E & Tech Media
Ph: +1 860 536 6677
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-4 to C-11, Hosiery Complex, Phase-II Extension, NOIDA-201305, Gautam Budh Nagar, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2018. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period Year	Newstand Price (₹)	You Pay (₹)	Overseas Digital	Overseas Print
Five	3600	2160	US\$ 20	—
Three	2160	1515	US\$ 15	—
One	720	575	US\$ 5	US\$ 100

Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com>



TECH FOCUS

34 Mobile Money: The Age of Digital Payments



FOR 3D PRINTER

Connect To The Page No. 105

www.rajguruelectronics.com

SMART WORLD

- 112 Li-Fi: Tomorrow's Connectivity with the Power of Light
- 114 Smart Technology to Improve Workforce Safety
- 116 Utilising Waste Plastic Bags to Design Tiles

RESEARCH & DEVELOPMENT

- 18 AGRICULTURE: Smart Agriculture and Agricultural Automation
- 24 PRODUCT DEVELOPMENT: Flex and Rigid-Flex PCBs
- 48 DESIGN: High Security Designs for Remote Controls
- 56 TEST & MEASUREMENT: Evolving Technologies and Test Devices for the Best 5G Experience
- 62 WIRELESS: Latest Wireless Technologies, Devices and Protocols for Secure Home Automation
- 71 INTERVIEW: Kuldeep Malik, Director - Corporate Sales India, MediaTek: "By the upcoming Diwali season, we may see AI become an integral part of budget smartphones"
- 72 INTERVIEW: Yunsup Lee, Co-founder and Chief Technology Officer, SiFive and Krste Asanovic, Co-founder and Chief Architect, SiFive: "This ISA spec and its compatibility suites will always be free"

EVERYTHING YOU WANTED TO KNOW ABOUT...

- 28 EMBEDDED: Clock Gating for the Internet of Things
- 42 DEFENCE ELECTRONICS: Optronic Sensors: Applications in Detonics (Part 6 of 6)
- 59 SENSORS: Sensors for Consumer Electronics
- 91 GREEN TECHNOLOGY: Defeating Pollution with the IoT: Solutions in India

SOFTWARE

- 68 IoT Software for Businesses and Development Tools for Programmers
- 70 Mobile Applications for the IoT and Home Automation

MAKE IN INDIA

- 98 MARKET SURVEY: Where India Stands in Electronics Manufacturing Services
- 104 PROJECT REPORT: How to Set Up a Business Unit for Crimped-Wire Connector Manufacturing
- 118 STARTUP ZONE: Crux LX: A Unified Communications Platform for Small and Medium Businesses
- 121 STARTUP ZONE: VAPCare: Medical Electronics to Improve ICUs
- 122 STARTUP ZONE: EcoFrost: Made in India Solar-Powered Cold Storage to Reduce Crop Spoilage

eSTYLE

- 126 FIRST LOOK: Some New Gadgets
- 130 BUYERS' GUIDE: Choosing the Best Wi-Fi Router for Your Home
- 132 ARTIFICIAL INTELLIGENCE: Potential Risks of Artificial Intelligence

DO-IT-YOURSELF



- 74 The IoT and Touch-Based Home Automation
- 78 Arduino-Controlled 12V Battery Charger
- 80 Picture Slideshow Using ESP32
- 82 Designing Wireless Gas Detection and Monitoring Systems
- 87 Accessing Graphical Desktop of Raspberry Pi Using SSH and VNC
- 89 Aircraft Signal Receiver Using RTL-SDR and DUMP1090

NEWS

- 12 Tech News
- 94 Industry News
- 123 New Products

Regulars

- 08 Feedback: Your Suggestions
- 10 Q&A: Things You Wanted to Know!